Alternative Index Policy

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December 28, 2023

The formulation for restless multi arm bandits is:

$$max\ E\ [\lim_{m\to\infty} \frac{1}{m} \sum_{n=1}^{n=m} \sum_{i=1}^{i=N} r^i(s_n^i, a_n^i)] \tag{1}$$

$$\text{subject to: } \sum_{i=1}^{i=N} a_n^i = M, \quad M < N \tag{2}$$

The Whittle Relaxation used is:

$$E\ [\lim_{m\to\infty} \frac{1}{m} \sum_{n=1}^{n=m} \sum_{i=1}^{i=N} a_n^i] = M \tag{3}$$

The relaxed LP formulation for restless multi arm bandits for N arms out of which M can be activated at a timestep can be written as:

$$max\ E\ [\lim_{m\to\infty} \frac{1}{m} \sum_{n=1}^{n=m} \sum_{i=1}^{i=N} r^i(s_n^i, a_n^i) + \lambda^*(\sum_{i=1}^{i=N} a_n^i - M)] \tag{4}$$

We propose an alternative index policy for restless multi arm bandits based on the difference of the Q values for the state (k) in which the arm is present in i.e.

$$\delta = Q_{\lambda^*}(k, 1) - Q_{\lambda^*}(k, 0) \tag{5}$$

Where λ^* is the common subsidy for passivity learnt by gradient ascent in the dual space:

$$\lambda_{n+1}^* = \lambda_n^* + \beta(n)(\sum_{i=1}^{i=N} a_n^i - M) \tag{6}$$

The Q values are updated according to the RVI Q-Learning algorithm as:

$$Q_{n+1}(s_n, a_n) \leftarrow (1 - \alpha(n))Q_n(s_n, a_n) + \alpha(n)((1 - a_n)(r_o(s_n) + \lambda_n^*) + a_n r_1(s_n) + \max_{v\epsilon\{0,1\}} Q_n(s_{n+1}, v) - \frac{1}{2d} \sum_{k\in S} Q_n(k, 0) + Q_n(k, 1)) \tag{7}$$

By using a common subsidy for passivity the problem complexity is reduced as we don't have to maintain and update Lagrangian multiplier for every arm (which also acts as the Whittle Index) instead we just have to learn the common subsidy

The alternative indexes are learnt offline using the following algorithm:

Once the alternative indexes and Q values are learnt they can be used online to select top M arms to be pulled

Algorithm 1 An Alternative Index Policy for Restless Multi Arm Bandits

Initialize λ^*, Q(s,a) for all states and actions, ϵ=1.
for n = 1: n_{end} **do**
  Choose action a_n^i for each arm i in an ϵ-greedy fashion
  Update s_{n+1}^i and reward r_n^i from s_n^i and a_n^i for every arm i
  Update $\alpha(n)$ and $\beta(n)$ as: $\alpha(n) = \frac{1}{\lceil n \backslash 5000 \rceil + 1}$ $\beta(n) = \frac{1}{\lceil nlog(n) \backslash 5000 \rceil + 1}$
  Update (s_n^i,a_n^i) Q-values for each arm i as:
  $Q_{n+1}(s_n, a_n) \leftarrow (1-\alpha(n))Q_n(s_n, a_n) + \alpha(n)((1-a_n)(r_o(s_n)+\lambda_n^*) + a_n r_1(s_n) + \max_{v \epsilon \{0,1\}} Q_n(s_{n+1}, v) - \frac{1}{2d} \sum_{k \in S} Q_n(k, 0) + Q_n(k, 1))$
  Update common subsidy for passivity for all arms λ^* as: $\lambda_{n+1}^* = \lambda_n^* + \beta(n)(\sum_{i=1}^{i=N} a_n^i - M)$
  Decrement ϵ as $\epsilon_{n+1} = max(0.01, \epsilon_n * 0.99)$
end for
Calculate the new index for each arm i where the current state of arm is represented by k:
  $\delta_{n+1}(k) = Q_{\lambda^*}(k, 1) - Q_{\lambda^*}(k, 0)$

1 Example with Circulant Dynamics

The example has four states and the dynamics are circulant: when an arm is passive (u = 0), resp. active (u = 1), the state evolves according to the transition probability matrices

$$P_0 = \begin{bmatrix} 1/2 & 0 & 0 & 1/2 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \; and \; P_1 = P_0^T \tag{8}$$

respectively. The rewards do not depend on the action and are given by $r(1,0) = r(1,1) = -1, r(2,0) = r(2,1) = 0, r(3,0) = r(3,1) = 0$, and $r(4,0) = r(4,1) = 1$. Intuitively, there is a preference to activate an arm when the arm is in state 3. Indeed, the exact values of the Whittle indices, calculated are as follows: $\lambda_1 = -1/2, \lambda_2 = 1/2, \lambda_3 = 1$, and $\lambda_4 = -1$, which give priority to state 3. Consider a scenario with N = 100 arms, out of which M = 20 are active at each time. We initialize our algorithm with $\delta_i = 0$ and $Q(i,u) = r(i,u), \forall i \in S$

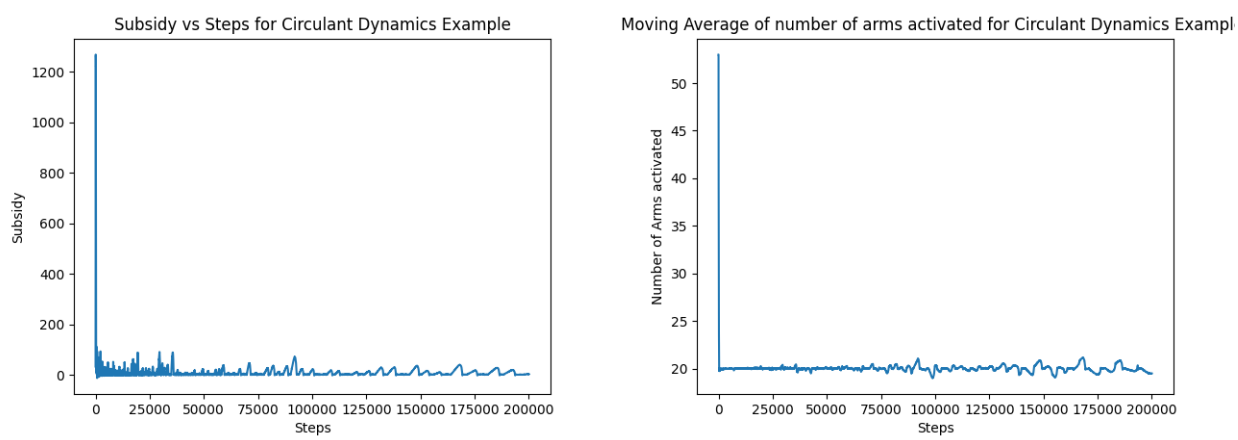

Figure 1: Above two plots are during the training phase

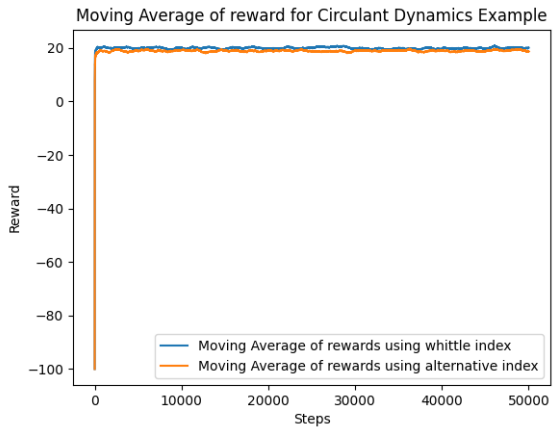

Figure 2: Above plot compares performances during online selection

2 Example with Restart

Now we consider an example where the active action forces an arm to restart from some state. Specifically, we consider an example with 5 states, where in the passive mode (u = 0) an arm has tendency to go up the state space whereas in the active mode (u = 1) the arm restarts from state 1 with probability 1, i.e.,

$$P_0 = \begin{bmatrix} 1/10 & 9/10 & 0 & 0 & 0 \\ 1/10 & 0 & 9/10 & 0 & 0 \\ 1/10 & 0 & 0 & 9/10 & 0 \\ 1/10 & 0 & 0 & 0 & 9/10 \\ 1/10 & 0 & 0 & 0 & 9/10 \end{bmatrix} \; and \; P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{9}$$

The rewards in the passive mode are given by $r(k, 0) = a^k$ (in our numerical experiments, we have taken a = 0.9) and the rewards in the active mode are all zero. As in the previous example, we consider the scenario with N = 100 arms out of which M = 20 are active at each time step. the exact values of the Whittle indices are given by: $\lambda_1 = -0.9, \lambda_2 = -0.73, \lambda_3 = -0.5, \lambda_4 = -0.26$ and $\lambda_5 = -0.01$. We initialize our algorithm with $\delta_i = 0$ and $Q(i, u) = r(i, u), \forall i \in S$

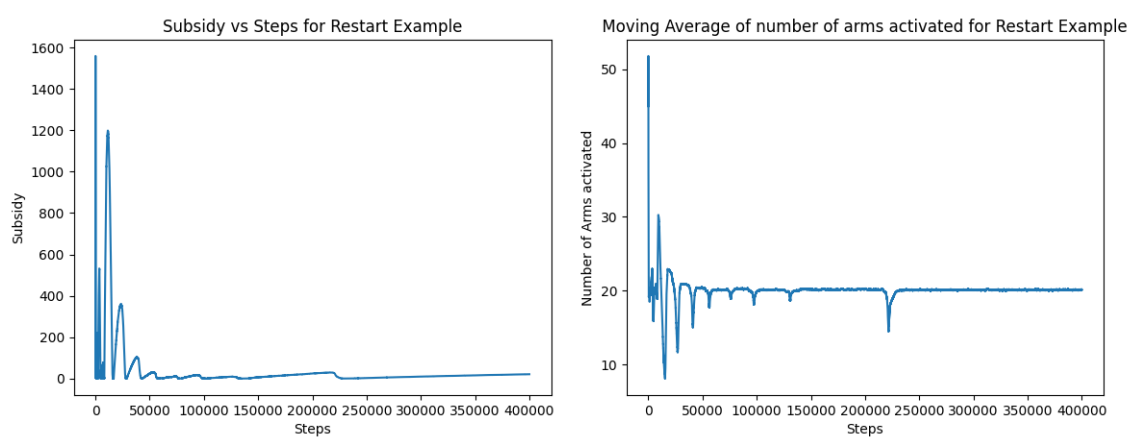

Figure 3: Above two plots are during the training phase

Figure 4: Above plot compares performances during online selection

3 Example with combined arms of both the above cases

In this case we take 50 arms with properties as mentioned in the example with restart and 50 arms with properties as mentioned in the example with circulant dynamics. We learn only one common subsidy for passivity for all 100 arms. The results show that even with this set of arms the mean number of arms getting pulled does converge to 20 as desired. The performance of online selection of 20 arms according to the new index is almost same as the performance of online selection of 20 arms according to the exact Whittle indexes stated in the above examples.
This shows the benefit of using a single learnt common subsidy of passivity. Even if the homogeneity of arms is taken advantage of when computing their Whittle indexes, it would still have to be done for the two different sets of arms separately.

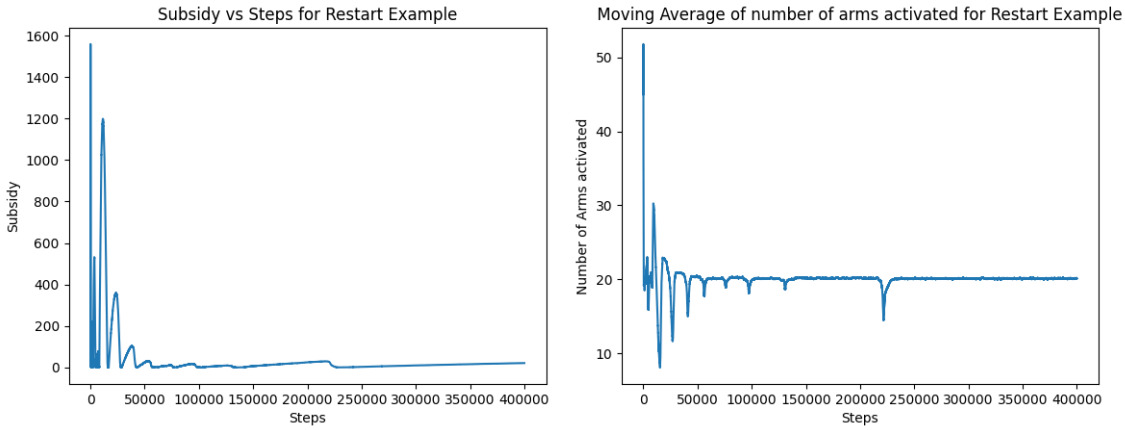

Figure 5: Above two plots are during the training phase

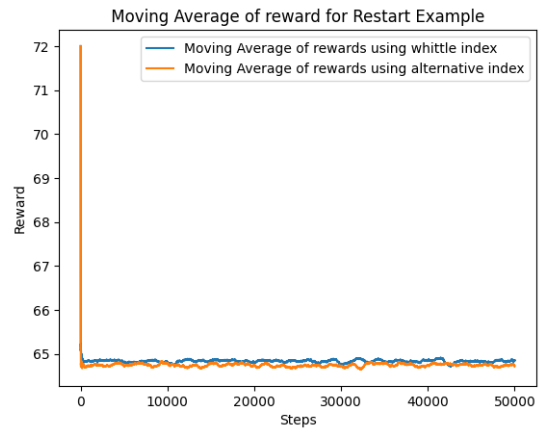

Figure 6: Above plot compares performances during online selection

4 Non Homogeneous Arms with Restart

In this example we take 5 arms as described in section 2 but all with different transition probabilities, i.e. non homogeneous.

$$P_0 = \begin{bmatrix} 1-x & x & 0 & 0 & 0 \\ 1-x & 0 & x & 0 & 0 \\ 1-x & 0 & 0 & x & 0 \\ 1-x & 0 & 0 & 0 & x \\ 1-x & 0 & 0 & 0 & x \end{bmatrix} \; and \; P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{10}$$

here x takes values from $\{0.5,0.6,0.7,0.8,0.9\}$ corresponding to the 5 arms. From the 5 arms we want to pull 1 arms i.e. N=5 and M=1. The rewards in the passive mode are given by $r(k,0) = x^k$ and the rewards in the active mode are all zero.

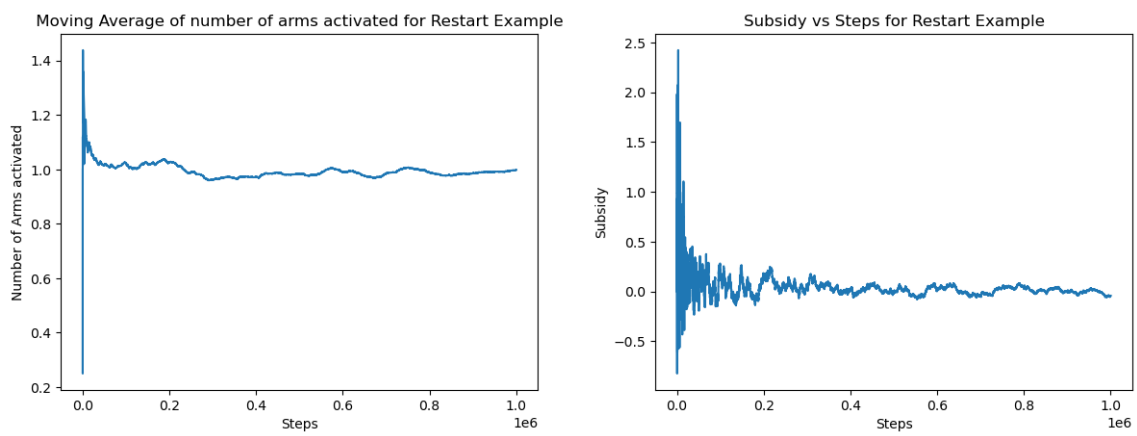

Figure 7: During training phase

We calculate the Whittle indices, for comparison of performance during online selection, for all the states of each arm using the Q-learning for Whittle Index (QWI) Algorithm. Below are the graphs showing the convergence of Whittle index for all the 5 arms. The Y axis is Whittle index and the X axis is number of steps.

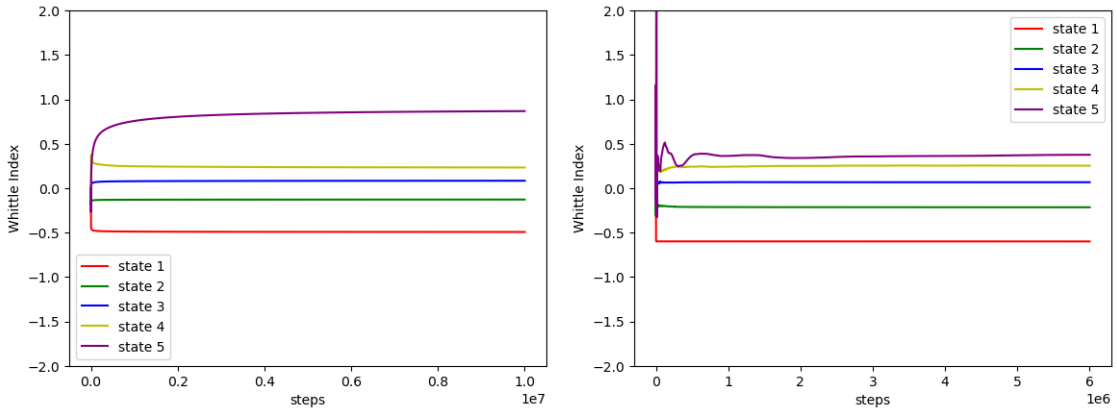

Figure 8: Above two plots are for arm 1 and arm 2

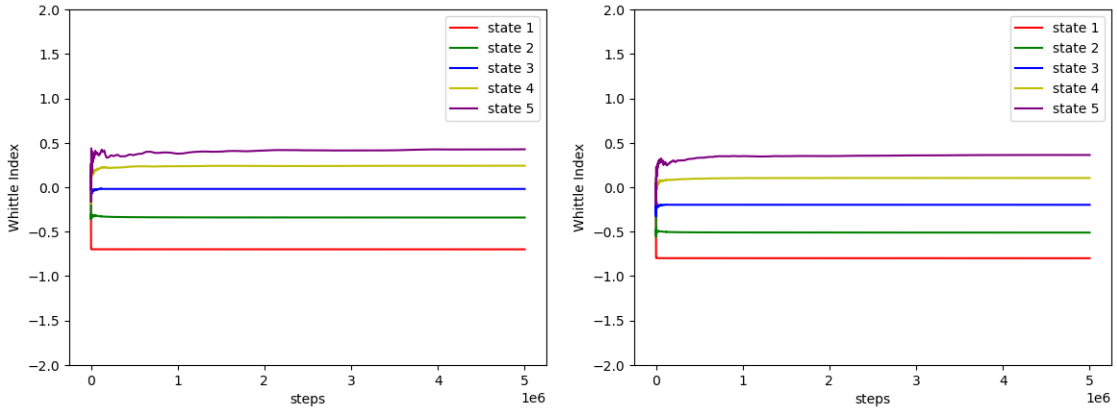

Figure 9: Above two plots are for arm 3 and arm 4

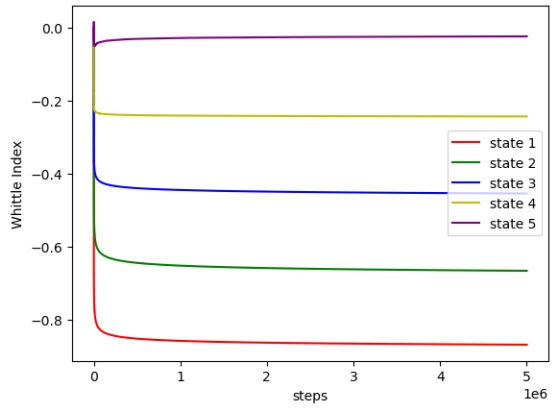

Figure 10: Above plot is for arm 5

Once the whittle indexes are calculated we compare the rewards obtained when choosing arms according to the proposed alternative index vs choosing arms according to the calculated whittle indexes.

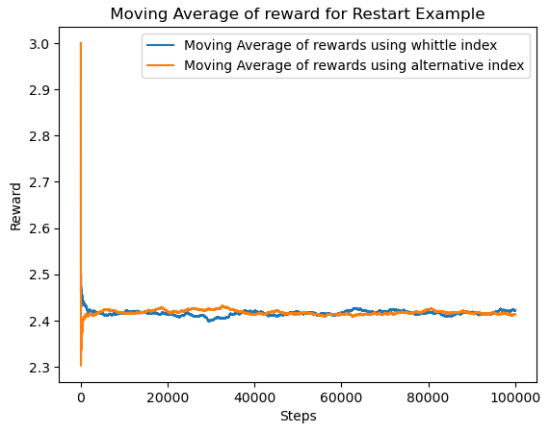

Figure 11: Above plot compares performances during online selection

Time Analysis:

- Time taken to calculate the whittle index for one type of arm is $\sim$ 2 mins
- Time taken to calculate new index for all arms is $\sim$ 1.5 mins

5 Multiple Non Homogeneous Restart Arms

In this case, we take a set of 50 arms: 10 restart arms with x=0.5, 10 restart arms with x=0.6, 10 restart arms with x=0.7, 10 restart arms with x=0.8, and 10 restart arms with x=0.9. For all the 50 arms we still learn just one common subsidy of passivity and for comparison during online selection use the Whittle indices calculated in section 4.

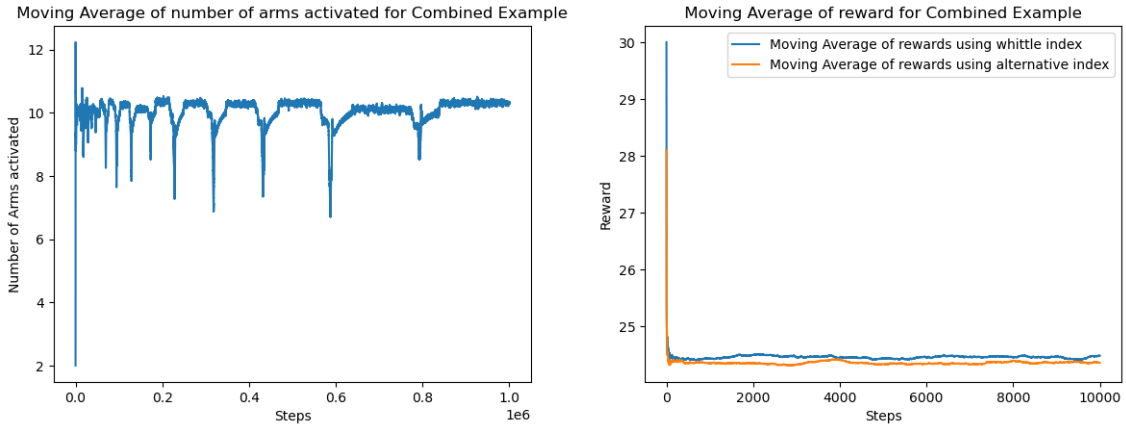

Figure 1 above shows convergence to choosing 10 arms during training phase. Figure 2 shows the comparison of rewards during online selection

6 Using DQN instead of tabular Q learning for alternative indices

Till now we have used tabular Q learning for learning Q values and ultimately the alternative indices. Now, we use the Deep Q Learning Network (DQN) for learning the Q values. This will especially be better in cases where the state space is large. The equation for Q target used in training DQN is:

$$Q_{target}(s_n, a_n) \leftarrow ((1-a_n)(r_o(s_n)+\lambda_n^*) + a_n r_1(s_n) + \max_{v\epsilon\{0,1\}} Q_n(s_{n+1}, v) - \frac{1}{2d}\sum_{k\in S} Q_n(k,0) + Q_n(k,1)) \tag{11}$$

The alternative indexes are learnt offline using the following algorithm: Below are the results obtained in

Algorithm 2 An Alternative Index Policy for Restless Multi Arm Bandits using DQN

Initialize λ^*, DQN, Batch Size,ϵ=1.
for n = 1: n_{end} **do**
    Update $\beta(n)$ as: $\beta(n) = \frac{1}{\lceil nlog(n)\backslash 5000\rceil + 1}$
    for arm $i \in N$ **do**
        Choose action a_n^i for each arm i in an ϵ-greedy fashion
        Update s_{n+1}^i and reward r_n^i from s_n^i and a_n^i for every arm i
        Store $(s_n^i, a_n^i, r_n^i, s_{n+1}^i, \lambda_n^*)$ in experience replay buffer
        Train DQN on one batch of $(s_n^i, a_n^i, r_n^i, s_{n+1}^i, \lambda_n^*)$ tuples from experience replay buffer
    end for
    Update the DQN target model
    Update common subsidy for passivity for all arms λ^* as: $\lambda_{n+1}^* = \lambda_n^* + \beta(n)(\sum_{i=1}^{i=N} a_n^i - M)$
    Decrement ϵ as $\epsilon_{n+1} = max(0.01, \epsilon_n * 0.99)$
end for
Calculate the new index for every state s of each arm i by:
    $\delta(s) = Q_{\lambda^*}(s,1) - Q_{\lambda^*}(s,0)$

the case of using N=10 restart arms as described in section 2, out of which M=2 have to be activated. Batch Size of 32 is used.

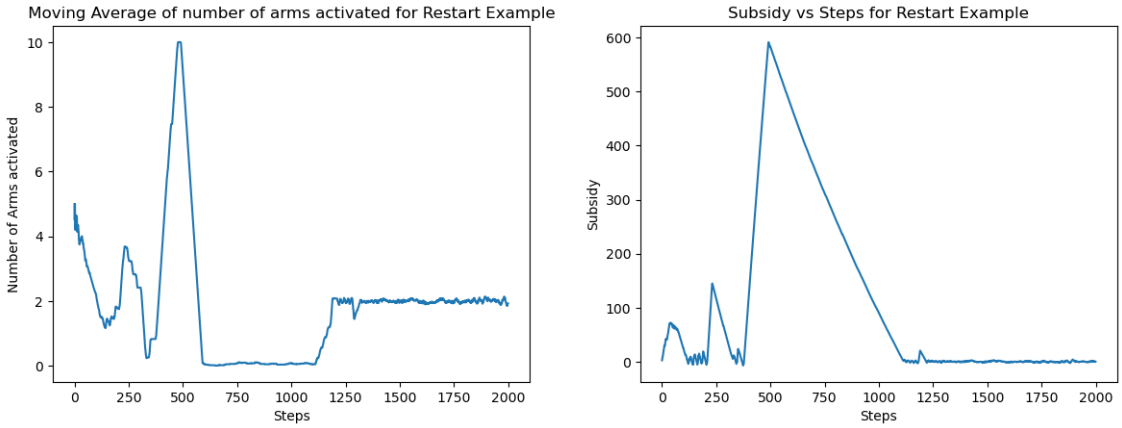

Figure 12: Above two plots are during the training phase

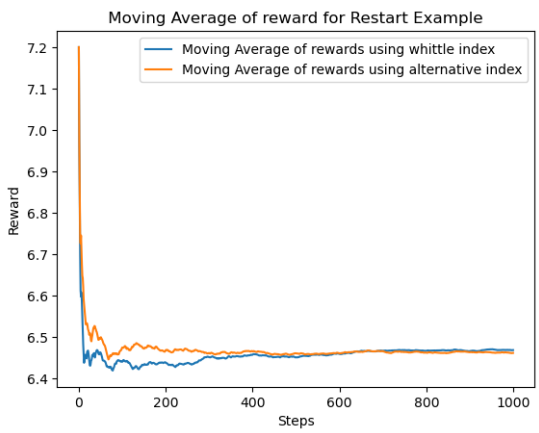

Figure 13: Above plot compares performances during online selection

7 Using DQN in non homogenous restart bandits case

In this case, we use a two-dimensional state space. The first dimension is used to give information about the type of arm and the second dimension is the state of the arm. The first dimension used are 0,1,2,3,4 for x=0.5, x=0.6, x=0.7, x=0.8 and x=0.9 respectively. So for example, if it is the arm with x=0.5 and it's state is 2 the state will be [0,2].

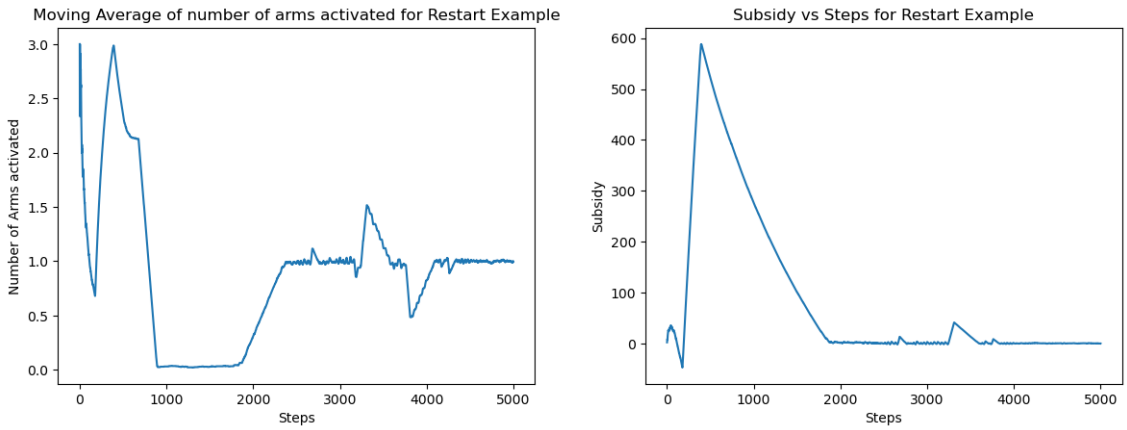

Figure 14: Above two plots are during the training phase

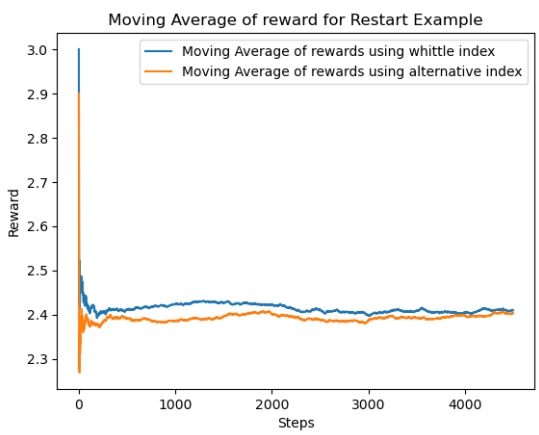

Figure 15: Above plot compares performances during online selection

8 Non Whittle Indexable Problem

Here we check the performance of our index in a Non Whittle Indexable Problem.
Whittle Indexability: As the passivity subsidy λ increases from $-\infty$ to $+\infty$ the set of project states where it is optimal to rest the project increases monotonically from the empty set to the full project state space

$$P_0 = \begin{bmatrix} 0.005 & 0.793 & 0.202 \\ 0.027 & 558 & 0.415 \\ 0.736 & 0.249 & 0.015 \end{bmatrix} \; and \; P_1 = \begin{bmatrix} 0.718 & 0.254 & 0.028 \\ 0.347 & 0.097 & 0.556 \\ 0.015 & 0.956 & 0.029 \end{bmatrix} \tag{12}$$

$$\text{reward matrix: } [[0, 0.699], [0, 0.362], [0, 0.715]] \tag{13}$$

Many times the Whittle index technique is empirically applied even if the problem is not indexable. Hence even we calculate the Whittle Indices in this case and compare the performance of our indices against it.

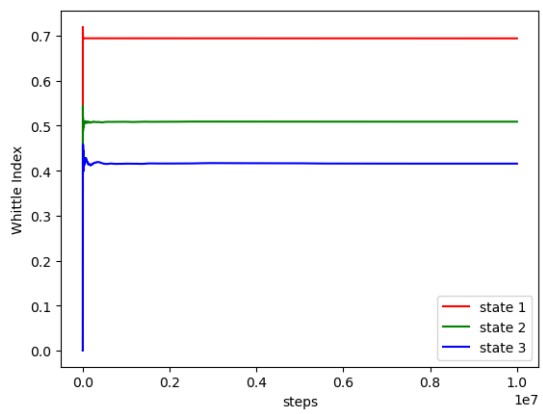

Figure 16: Whittle Indices calculated using QWI

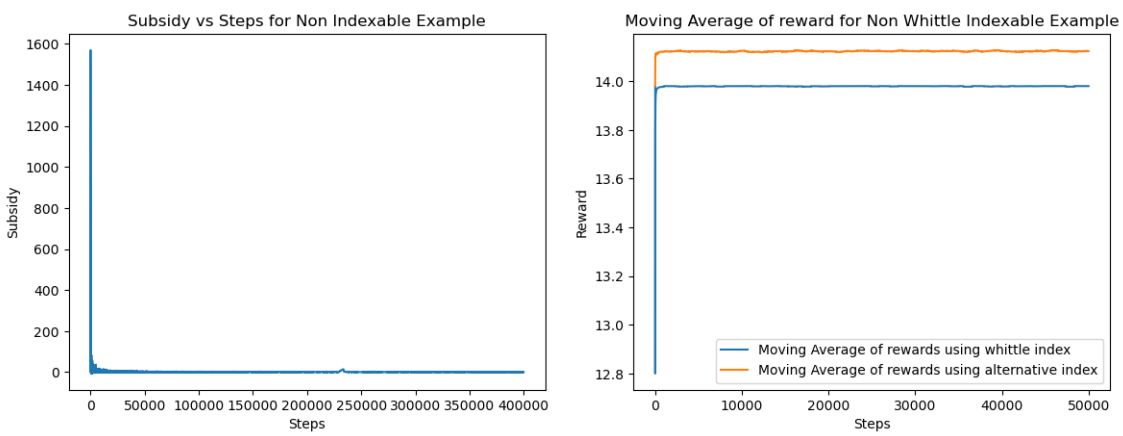

Figure 17: Above plot compares performances during online selection

As seen the new indices result in a higher reward as compared to the whittle indices, thus showing that the new index can also be used in Non Whittle Indexable cases as well.